

UNIT-5 | Lecture 2

Evaluation of Relational Algebra — Cost & Execution

Coverage: Hour 2

Learning Objectives

Understand the cost of evaluating relational algebra operations.

Identify important factors affecting query execution cost.

Query Evaluation Cost

The cost of a query can be measured using **disk I/O, CPU processing, memory usage, and intermediate-result size**.

Disk I/O is often a major factor because reading/writing data pages can be expensive.

Selection Evaluation

A selection can use a **linear scan** when no useful index exists.

An **index** may reduce the number of pages that must be read when the search condition is selective and the index is useful.

Projection Evaluation

Projection removes unwanted attributes.

Duplicate elimination may require sorting or hashing, which adds cost.

Join Evaluation

Join is often one of the most expensive relational operations.

The choice of join strategy depends on relation sizes, available indexes, memory, ordering, and join condition.

Cost-Saving Principles

Push selections toward the base relations.

Project away unnecessary attributes early when safe.

Choose efficient join order and algorithms.

Avoid generating unnecessarily large Cartesian products.

Key Takeaway

Query evaluation cost depends heavily on I/O, intermediate-result size, memory, and the algorithms chosen for each operator.